CSIT970 Security Essentials

Week 3 – Cryptographic Issues

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Cryptography

- It is an inevitable line of defence for many systems and applications.
- It does not solve all the problems but many.
- Essential knowledge IT personnel should gain.
  - It is impossible to know cryptography in every detail,
  - but it's important to understand it (concept-application) correctly.

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Timeline in Cryptography

Cryptography ≠ Encryption + Decryption

1976

Symmetric Key Encryption

Key Exchange
Public Key Encryption
Digital Signature
Cryptographic Hash function
Message Authentication Code

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Taxonomy of Cryptography

- Cryptography
  - Key-Based Cryptography
    - Symmetric
      - Symmetric Key Encryption
      - Message Authentication Code
    - Asymmetric
      - Key Exchange
      - Public Key Encryption
      - Digital Signature
  - Keyless Cryptography
    - Cryptographic Hash function

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Taxonomy of Cryptography

Suppose a cryptography is used between Alice and Bob.

- If this cryptography requires Alice and Bob to share **the same secret key** first, it is called symmetric cryptography
- Otherwise, it is called asymmetric cryptography

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Key in Cryptography

In Cryptography, a key is usually represented with n-bit string

Once the key is known by the adversary, it can immediately break the scheme.

When the n-bit key is randomly chosen, the probability of guessing it is $\frac{1}{2^n}$.

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Key Size?

- Calculation: To find a random key by brute-force
  - n: key size (in bits)
  - c: CPU speed (Usually GHz/s) → Clock cycles (cc) per second
  - Time taking for brute-force attack: $t = 2^n/c$ (Assumption: it takes one clock cycle (cc) to perform one instance of search)
  - Example

    60-bit key searched by a computer with i5 3GHz processor, i.e. $c = 3 * 10^9\ cc/s$

    $t = 2^{60}/(3*10^9)\ s$ = 384,307,168 seconds = 160,752 hours = 4,448 days = 12 years

    60-bit key searched by a super computer with $c = 2*10^{16}\ cc/s$

    $t = 2^{60}/(2*10^{16})\ s$ = 58 seconds!

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Key Size?

- Then, how about other key size?
  - ➢Example

    128-bit key, searched by a super computer with $c = 2*10^{16}$ cc/s

    $t = 2^{128}/(2*10^{16})$ s = 2^{49} (= 10^{15}) years! (Compare this number with any large number in https://en.wikipedia.org/wiki/Large_numbers)

    Hence n=128 is very safe
- Due to limitations from fundamental physics, there is no expectation that any digital computer (or combination) will be capable of breaking 256-bit random string via a brute-force attack.

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Symmetric (Key) Encryption

- A sender and a receiver use the same secret key *k*.
  - *C = Enc(k, M)* *C*: *Ciphertext;* *M: Plaintext (or Message)*
  - *M =Dec(k, C)*
- Stream ciphers
  - Basically, each bit of plaintext is XOR-ed with a key stream.
  - It is fast and often used as "an encryption scheme on the fly".
- Block ciphers (such as AES)
  - Plaintext is divided into blocks of pre-defined size (64 bits, 128 bits etc.) and each block is encrypted using a symmetric cipher called "block cipher".

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Confidentiality (Secrecy) of Encryption

- Confidentiality (secrecy)
  - We assume that the plaintext that Alice sent to Bob is unknown to the adversary.
  - However, the adversary does have the ciphertext.
  - The adversary should not get anything useful information about the plaintext from the ciphertext.
    - ✓ This is a very information definition from the theoretical cryptographer's point of view, though.

Adversary

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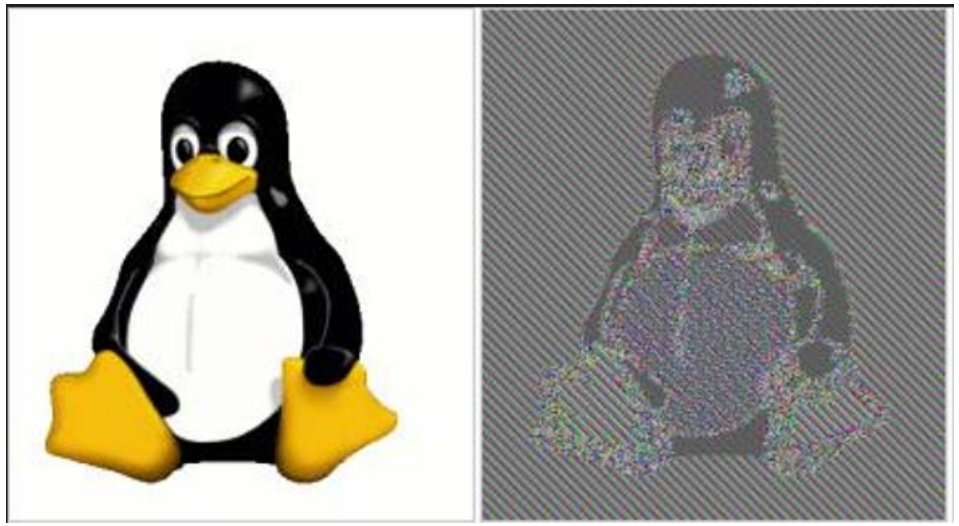
Confidentiality (Secrecy) in Cryptography

- Look at the following picture. Do you think this is good encryption?



plaintext ciphertext

Adversary

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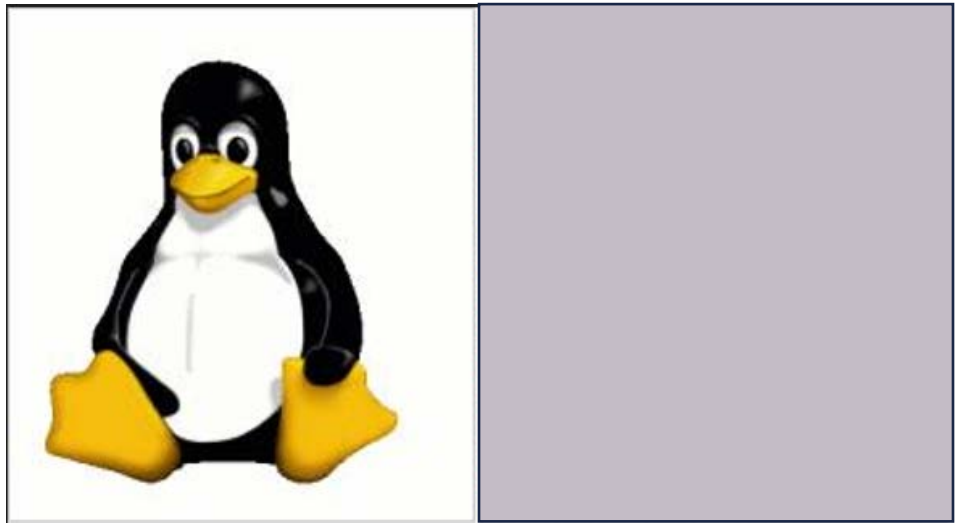
Confidentiality (Secrecy) in Cryptography

- We want something like this…



plaintext ciphertext

Adversary

Looks like easy, but many encryption algorithms fail to achieve this!

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Public Key Encryption

- Symmetric key encryption is efficient, but Alice and Bob might not have a shared secret key before.
- Public key encryption aims to achieve confidentiality between two users who do *not share a secret in advance*.
- The concept of public key encryption was proposed by Diffie and Hellman (in 1976) and the first public key encryption scheme, which is known as RSA, was introduced by **R**ivest, **S**hamir and **A**dleman.

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Public Key Encryption

- Notation: *pk*= public key, *sk*=secret key: *pk* is a key that can be published: *sk* is a secret key that the owner must keep secret
- In a public key encryption scheme, each user is assumed to generate a key pair (*pk*, *sk*).
  - The sender uses *pk* to encrypt her (plaintext) message *M*: *C* =Enc(*pk*, *M*)
  - The receiver uses *sk* to decrypt C: *M* = Dec(*sk*, *C*)
- Security: The most basic security property is that from *pk*, no one can come up with *sk*. Also, without having *sk*, no one will be able to get *M* from *C*. Usually these properties are guaranteed by the computationally hard problems. For this, *sk* and *pk* must be large!

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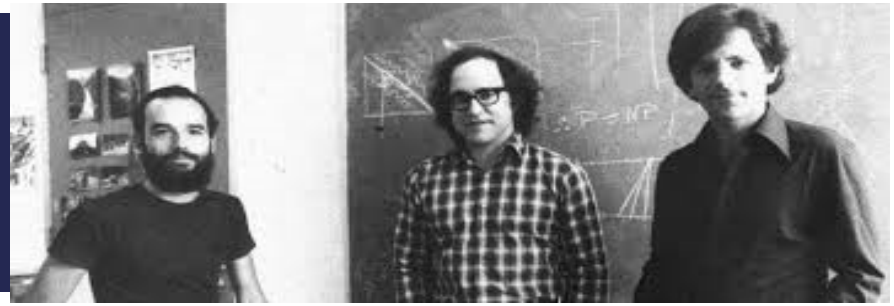
Public Key Encryption



- Example: RSA
  - $pk = (N,e)$, where $N=pq$, p & q primes; $sk=d$, where $ed=1 \bmod (p-1)(q-1)$
    - ✓ Hard to get d from (N, e): To get d, one needs to calculate $(p-1)(q-1)$. Without knowing the prime factors p and q, it seems hard to calculate $(p-1)(q-1)$.
    - ✓ Factoring N into p and q is known to be hard → Prime factorisation problem.
  - $\mathrm{Enc}(pk, M) = M^e \pmod N$
    - ✓ Hard to get M from C without knowing d: To get d, one needs to calculate $(p-1)(q-1)$. Again, factorisation problem.

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Integrity and Authentication in Cryptography

- Integrity
  - We assume that the message is **known** to the adversary.
  - The adversary wants to alter (modify) the message in transit.
  - The receiver of the message should detect something has been altered.
- Authentication
  - We assume that the message is **known** to the adversary.
  - The adversary wants to alter (modify) the message and claims that the message was originated from a specific person, say Alice.

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Cryptographic Hash Functions

- Cryptographic hash functions are applied to guarantee the integrity of data. It takes the arbitrary length inputs and outputs a fixed length value.
- In particular, *cryptographic* hash function H satisfy the following property:
  - Preimage Resistant: It must be one way. Given *v* such that H(d)= *v*, finding value of d is infeasible.
  - Collision Resistant: It is infeasible to find the two distinct d and d' such that

$$H(d) = H(d')$$

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Cryptographic Hash Functions

- It is often used as a component of other cryptographic algorithms.
  - For example, hash functions are used to compress a large data to a smaller one which can be taken as a message of digital signature for the efficiency purpose.
  - It also can be used to generated a deterministic random values for a given value.
- MD5, SHA1, SHA2 and SHA3 are cryptographic hash functions.
  - MD5 and SHA1 are not recommended any more for secure implementations. (They are broken!)

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Message Authentication codes

- Message Authentication Codes (MACs) are symmetric cryptography. A MAC, sometimes known as a tag, is a short piece of information used for authenticating a message.
  - t = MAC(k, M), where k is a key and t is called a "tag" or "MAC".
  - {0,1} = Verify(k, t, M)
  - Someone who does not know the key cannot compute a valid tag.
  - Someone who does not know the key cannot verify a tag.

- Producing MACs from hash functions
  - Using collision-free hash function H, compute t = H(k,M).

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Message Authentication codes

- When MACs are used between Alice and Bob, they need to have a secret key k first.
- Digital Signatures
  - ➢message integrity – the message has not been altered in transit,
  - ➢message authentication – the message was sent by Alice,
  - ➢*non-repudiation – Alice cannot claim she did not send the message.*
- MACs
  - ➢message integrity – the message has not been altered in transit,
  - ➢message authentication – the message was sent by Alice,
- A signature can only be generated by Alice (Signer), but a tag can be generated by both Alice and Bob.

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Digital Signature

- A signer has a key pair (pk,sk).
- In digital signature, the signer signs a message with her signing key and the resulting signature is verified using the signer's public key:
  - Signing key *sk* is private (secret).
  - Verification key *pk* is public.
- Digital Signature scheme
  - s = Sign(sk, M) s: <u>*Signature*</u>
  - {"Invalid (0)" or "Valid (1)"} = Verify(pk, M, s)

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Digital Signature

- The output of digital signature provides a guarantee of three important security properties:
  - ➢message integrity – the message has not been altered in transit,
  - ➢message authentication – the message was sent by Alice,
  - ➢*non-repudiation – Alice cannot claim she did not send the message.*

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Public Key Infrastructure (PKI)

- In public key cryptography, a public key looks like this:

30 82 01 0a 02 82 01 01 00 dd 9e e4 f6 88 b5 0d d7 a1 5f bc 25 6e cc 44 14 cf 34 e2 b5 73
09 1e e4 4b 12 52 38 95 36 1a c6 66 ed f0 c8 03 c9 b3 43 45 4e 0d 6a 92 4b 1b eb 94 60 5b
11 b9 15 79 b1 a5 f6 fc 5d bf a4 30 59 84 02 dd 3f 6d 21 6a 44 b7 18 1c 24 fc f5 02 2e 87 0c
20 3e c6 c5 b6 9f ad 16 1b 76 86 e9 73 9c 8d 31 60 3a a0 f0 2f da ad 8e f6 74 c9 81 d3 ea
f7 5d ab 5d bb 05 63 b0 78 55 ed 72 13 a4 42 43 72 23 73 c0 de 33 9b 44 5c 89 a9 8a 90 d1
99 be bc f7 21 21 5f fb 22 8a 5c 50 b9 69 7c dc 87 92 ed 79 56 ed 32 55 41 9f af 41 f6 da d4
70 88 e9 a3 41 1d 66 9f a6 98 d2 7e 5b a9 52 38 1c 56 b4 cc 45 62 72 0c c7 f7 ef 2c 47 0a
3b 1a 7a ac e7 ae a9 a8 1e 98 43 b0 58 56 e9 41 44 72 e6 da 67 1c d7 b6 f4 e6 b4 90 5c b5
0a 98 b3 23 0c e7 35 6d 10 14 73 0e 94 5d 7c 4e 0a 18 f4 05 20 67 9f 02 03 01 00 01

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Public Key Infrastructure (PKI)

- In public key cryptography, public keys need to authenticated.
  - In other words, the relationship between the user's identity and the public key the user has created should be verified.
  - If this is not verified, there is a possibility that ciphertexts, signatures and other public key cryptographic objects are all bogus. (Man-in-the Middle attack)
- A widely-accepted solution for this is to have Bob prove to the Certification Authority (CA) that his identity ID_{Bob} and public key pk_{Bob} are legitimate. As a token of this verification, Bob will get something from the CA.

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Public Key Infrastructure (PKI)

- That token is a digital signature generated as follows:
  - The CA signs Bob's ID with the public key that he has created:

$$s_{Bob} = \mathrm{Sign}(sk_{CA}, ID_{BOB} || pk_{Bob})$$

(That is, $ID_{BOB} || pk_{Bob}$ is regarded as a message input for the signature algorithm.)

  - Alice (or anyone) who has obtained CA's verification (public) key pk_{CA} can check whether s_{Bob} is valid by running the verification algorithm. If it is valid, she can trust Bob's public key.

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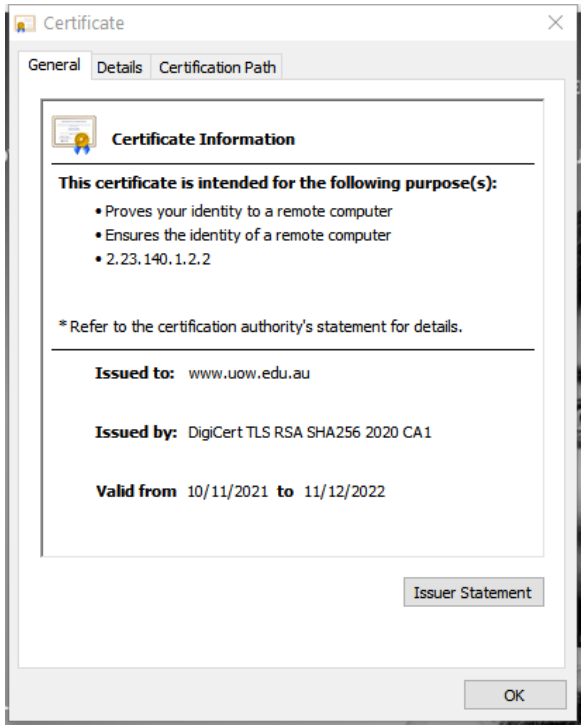
Digital Certificate

- Digital certificate is a digital file that contains the user's ID and the public key he/she created and CA's signature on them.
- Other extra pieces of information such as
  - ➢Type of Signature
  - ➢Signature serial number
  - ➢CA's information
  - ➢Validity (such as expiry date of certificate)

  are contained in certificates.



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Summary

- Each cryptography focuses on one specific security service.
- We might need to combine different cryptographies to have one security service protections.
- For example, Alice and Bob run Key Exchange to have a shared secret key before they run Symmetric-Key Encryption and Decryption.
- Asymmetric cryptography is much slower than symmetric cryptograpy.

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The Ability of Cryptography

- The security provided by cryptography means that no adversary can break a cryptographic scheme within a well-defined and specific environment. (Usually, in this environment, we assume that the adversary does not know the key.)
- In the real world, it is impossible to have security in the universal environment.
  - For example, all cryptography schemes are insecure if keys are known by the adversary.

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How to Bypass Cryptography Protections

- Passive Attacks (Implementation)
  - Ex) Side-channel attacks (Get the secret key)

- Active Attacks (Algorithm Substitution Attack)
  - Ex) Controlling key generation (Know the secret key)

  (Another issue is the key management and key theft)

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Side-Channel Attacks

In computer security, a side-channel attack is any attack based on information gained from the implementation of a computer system.

We can obtain the information of secret key via:

- Timing information (time cost of computing)
- Power consumption (off-peak, on-peak)
- Electromagnetic leaks
- Sound

(While cryptography assumes that the algorithms can be safely implemented)

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Side-Channel Attacks

- Example: RSA
  - $pk = (N,e)$, sk=d
  - Enc(pk, M) = M^e (mod N) =C
  - Dec(C, d)= C^d =M <u>(This is called exponentiation)</u>

Here, d is an n-bit string when decrypting the ciphertext.

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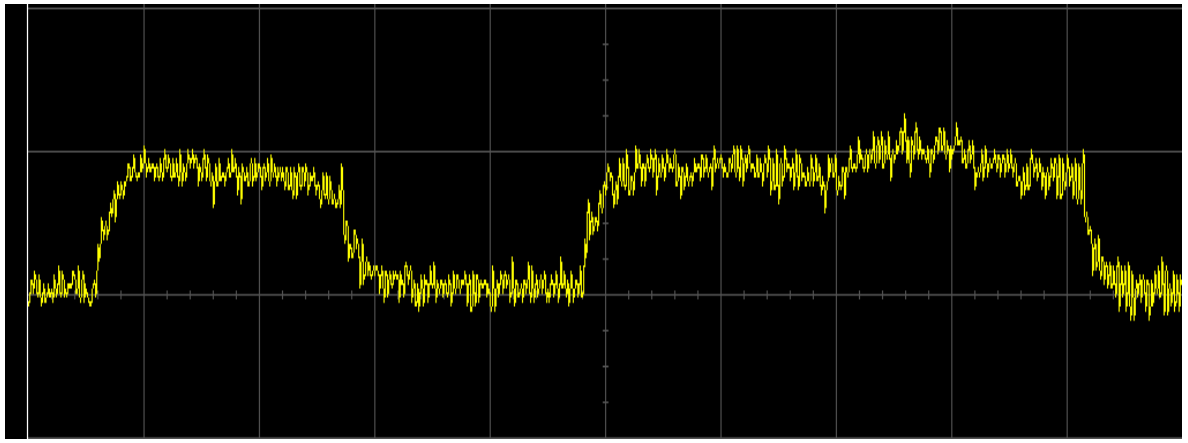
Side-Channel Attacks

ALGORITHM 1

LEFT-TO-RIGHT BINARY METHOD

Input:	X, N, $E = (e_{k-1}, \ldots, e_1, e_0)_2$
Output:	$X^E \bmod N$

```
R := 1;
for i = k − 1 downto 0 do
    R := R · R mod N;            — squaring
    if e_i = 1 then
        R := R · X mod N;        — multiplication
    end if
end for
return R
```



An attempt to decode RSA key bits using power analysis. The left peak represents the CPU power variations during the step of the algorithm without multiplication, the right (broader) peak – step with multiplication, allowing an attacker to read bits 0, 1. (From Wiki)

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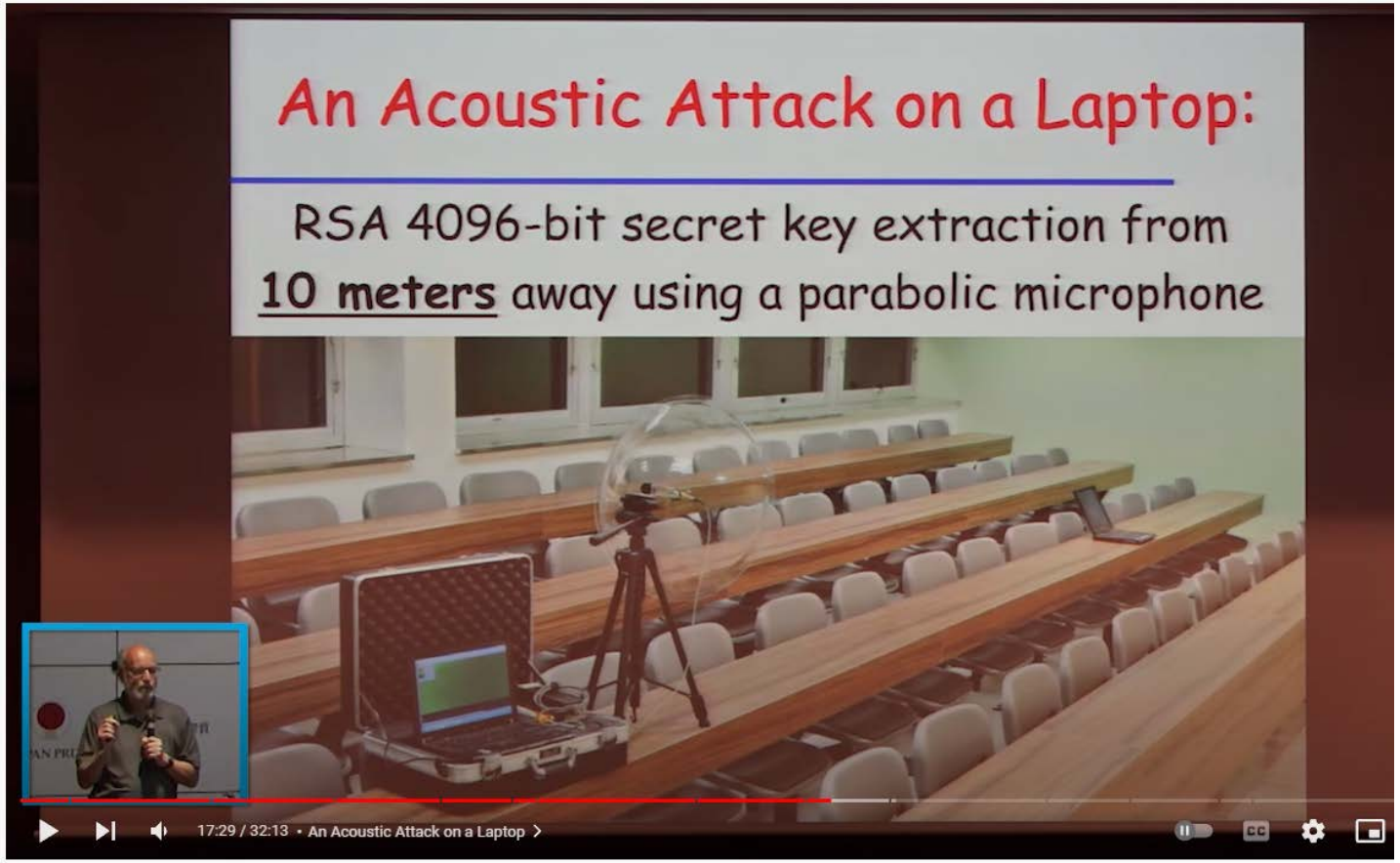
Side-Channel Attacks



2017 Japan Prize Commemorative Lecture: Dr. Adi Shamir

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Algorithm Substitution Attack

- Roughly, the ASA (Algorithm Substitution Attack) enables an adversary, often called “big brother”, to replace some parts of a legitimate cryptographic scheme or protocol with a modified one.
- This is related to “mass surveillance” we have learned.
- The ASA wants to let computation results subliminally leak the crucial information about a secret key or a confidential message to the adversary.

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Controlling Key Generation

- In cryptography, we run a key generation algorithm to generate a key *k* for symmetric cryptography or a key pair (*pk*, *sk*) for public key cryptography.

- The first step of the key algorithm is to generate a random *n*-bit string *S* first, which will be set as *k* or *sk*.

- It is hard to know whether the string *S* is indeed randomly chosen, because it is hard to judge whether the random algorithm is compromised or not (unless we can check the code).

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Controlling Key Generation

- For example, two very special bit strings

00000000000000

11111111111111

could be generated from a good random number generation algorithm.

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Controlling Key Generation

- Let `RandStr(n)` be the function that outputs a random n-bit string.
- We may want to call this function to generate a (random) secret key using the following function:

```
def KeyGen(n):
    A = RandStr(10)
    B = H(A)   #H is the hash function that outputs n-bit
    return H(A)
```

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Controlling Key Generation

- What is the problem of this function?
  - As long as we can correctly guess the output of RandStr(10) (A=**10-bit random string)**, we can recover the whole key from the KeyGen( ) function.
  - We only need to produce 2^{10}=1024 strings and try them one by one (This is called a brute-force attack).

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